# Diffused Photon NIR Tomography for Traumatic Brain Injury

Portable FD-fNIRs system concept for early TBI screening

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# Introduction

**3,800,000**

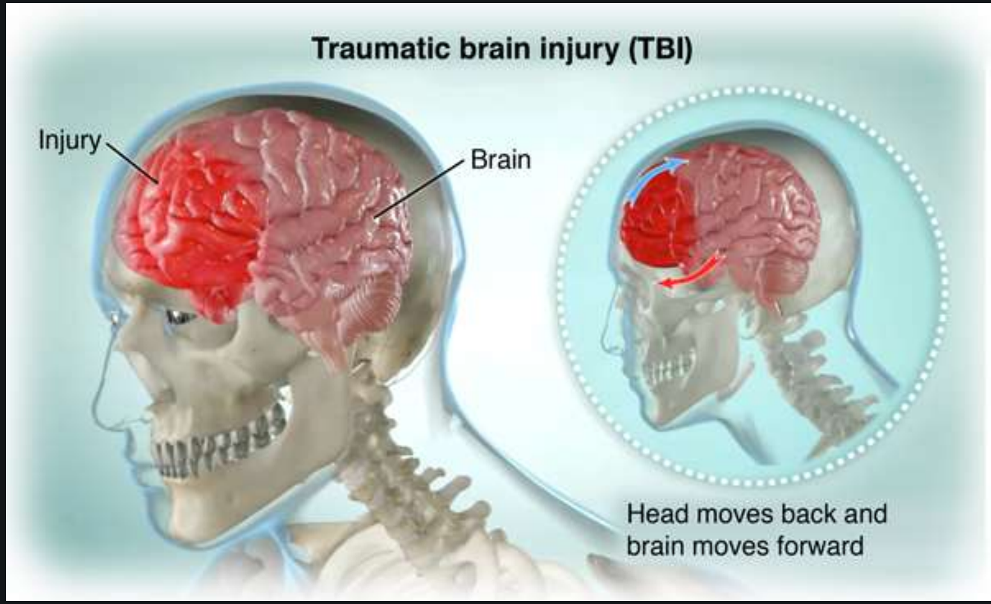
TBIs each year in the US

**10%**

Due to sports related injuries

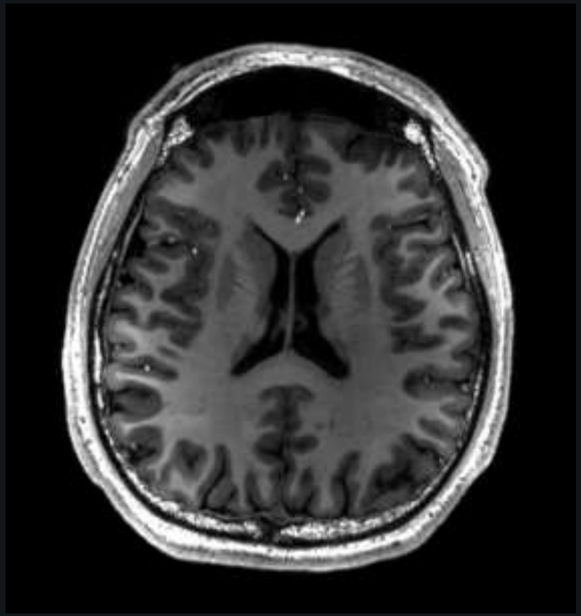
**492,000+**

U.S. service members diagnosed
with TBI from 2000-2023

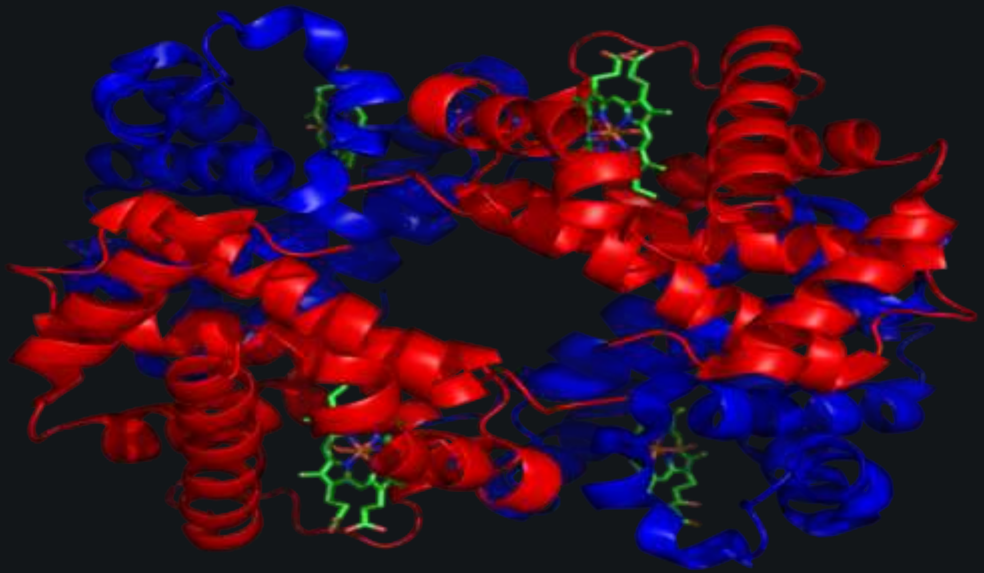


# Current Methods

- Magnetic Resonance Imaging (MRI)
- Computed Tomography (CAT Scan)
- X-Ray



# Significance of Hemoglobin

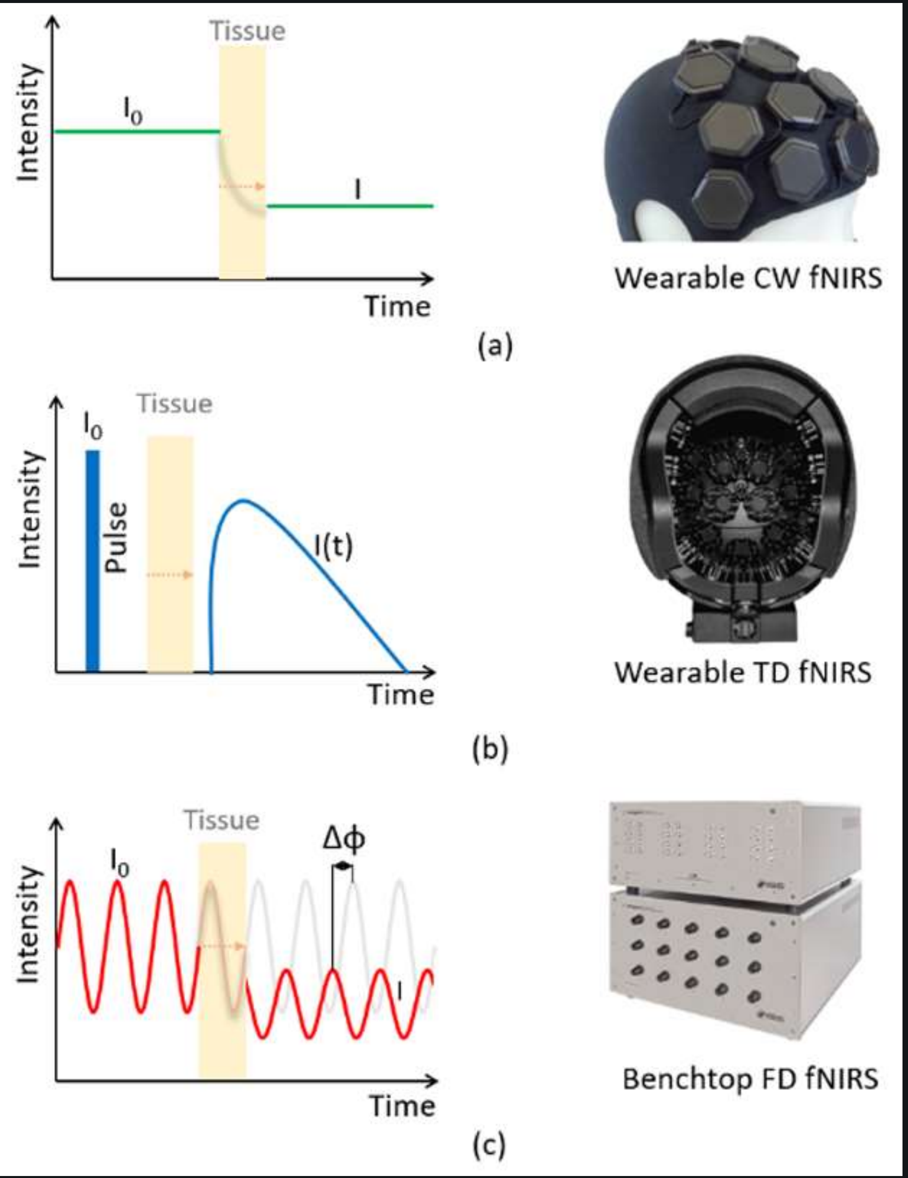


- Protein that delivers oxygen
- Absorption spectra
- In healthy brains
  - About even oxygenated to deoxygenated
- Impact damages small blood vessels
  - Oxygenated hemoglobin leaks out
  - Abnormal ratio

# Objectives

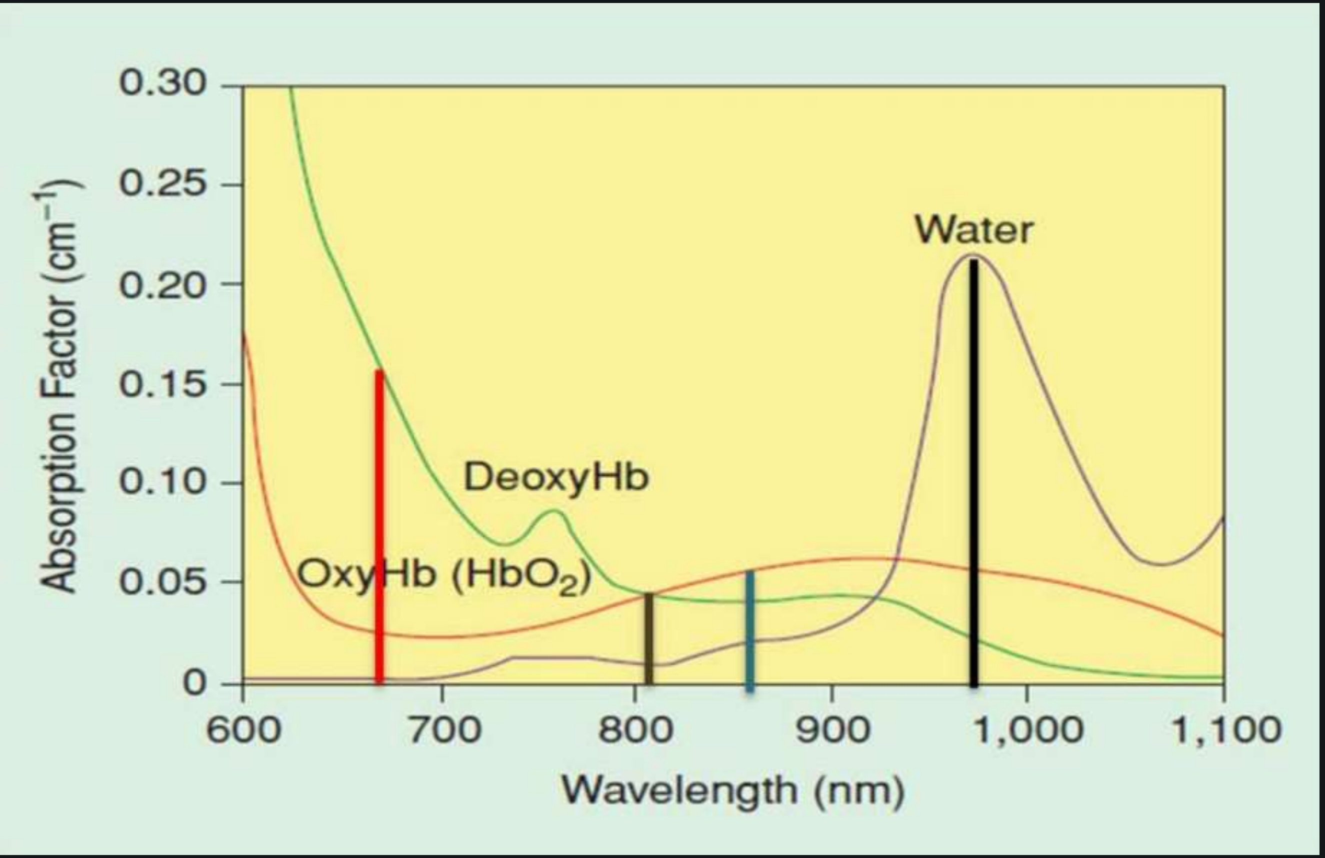
- Portable
- Fast
- Noninvasive
- Commercially available
- Implement
  - Transmitter
  - Receiver
- Calculations
- Imaging

# Why Frequency-Domain Functional Near-Infrared Spec



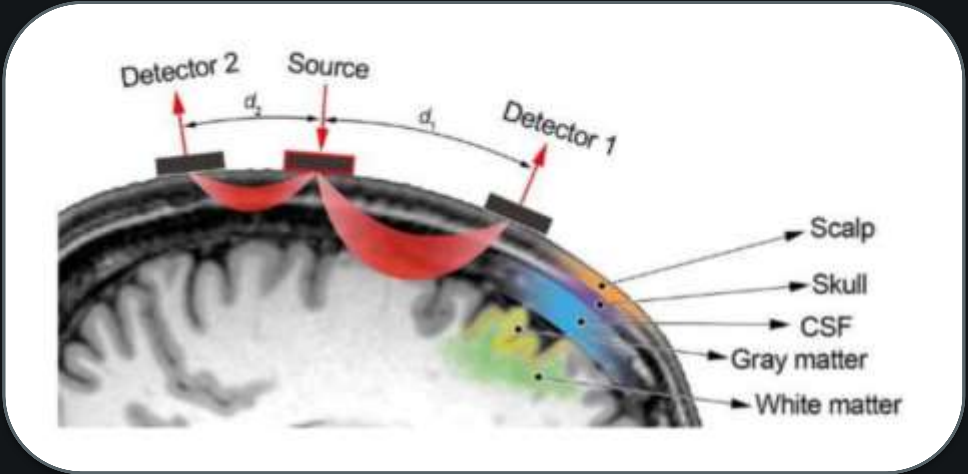

- Continuous Wave
  - Constant intensity
  - Measures change in intensity
- Time Domain
  - Short light pulses
  - Measures arrival time distributions
- Frequency Domain
  - Modulates emitted light intensity
  - Measures both intensity and phase

# Why Three Wavelengths?




- Absorption of oxygenated and deoxygenated hemoglobin
- Deoxygenated
  - 600-800nm
- Isosbestic point
  - Approximately 800nm
  - Absorbed identically
- Oxygenated
  - 850 nm
- Modified Beer-Lambert Law

# Approach for the Project




## FD-fNIR is a screening tool

Provides early information about possible brain injury and oxygenation changes; not intended to replace CT or MRI.

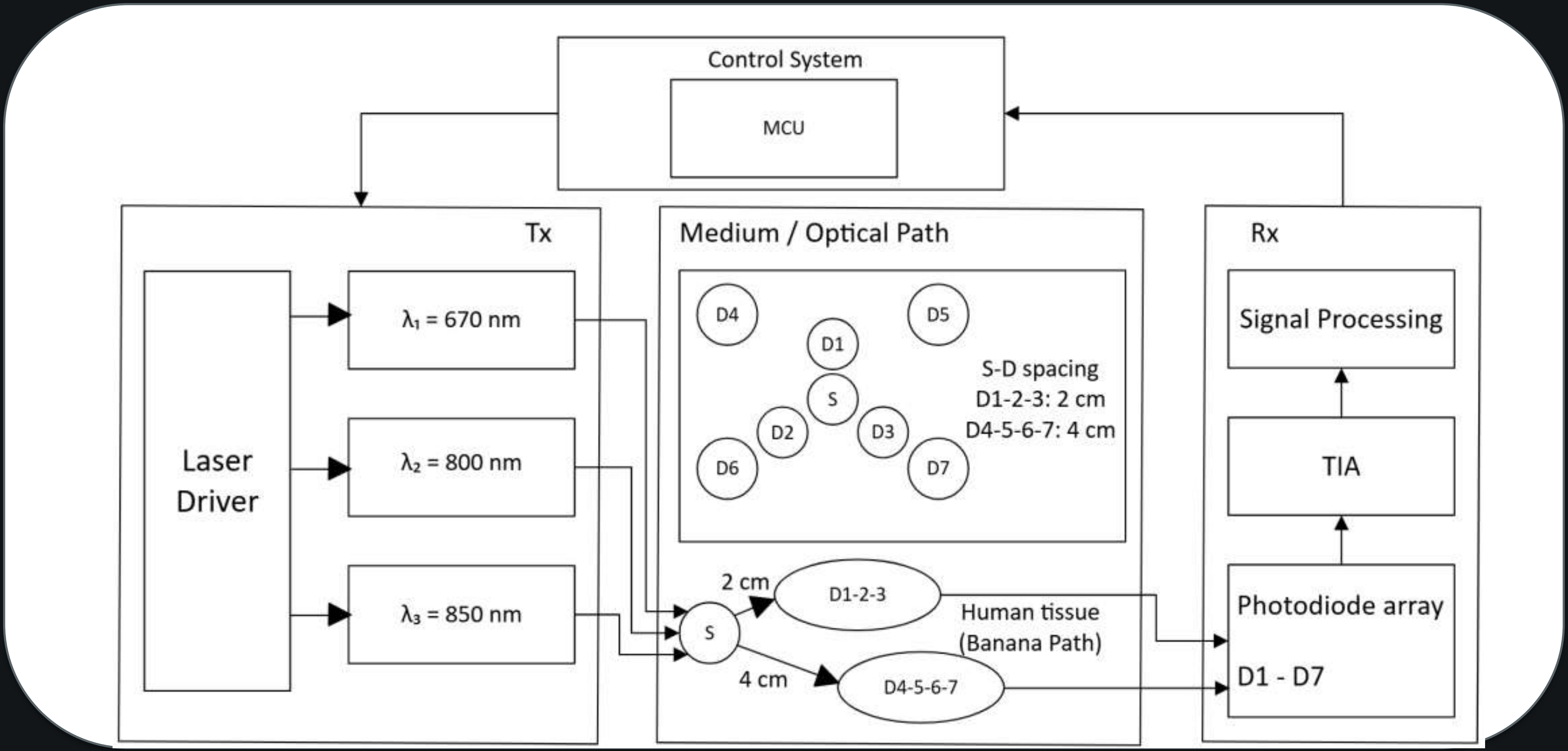
## Accuracy depends on signal quality

Sensor placement, hair, motion, and poor skin contact can affect the received signal.

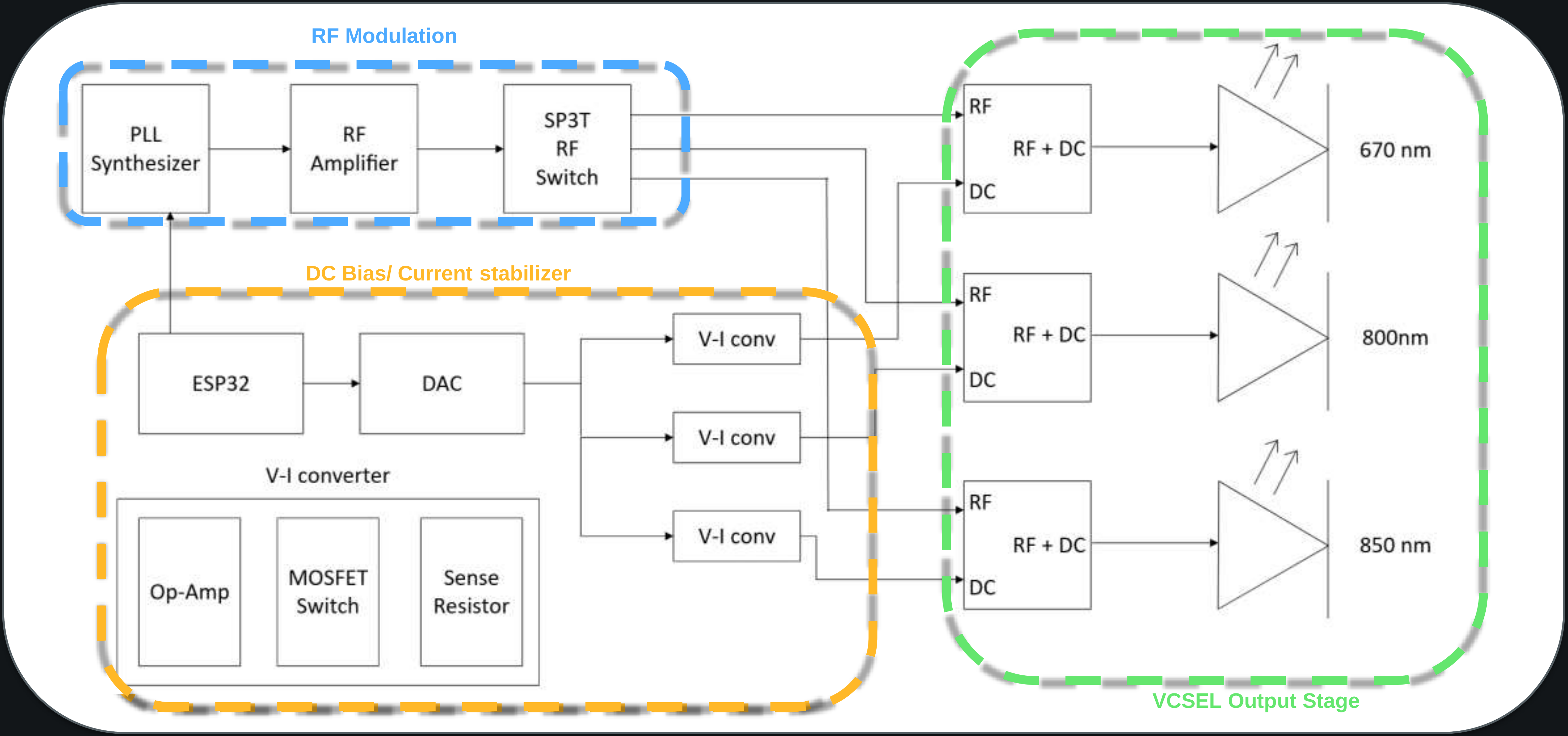
## Source-detector spacing is a tradeoff

Greater spacing improves penetration depth, but reduces received signal strength.

# General System Block Diagram

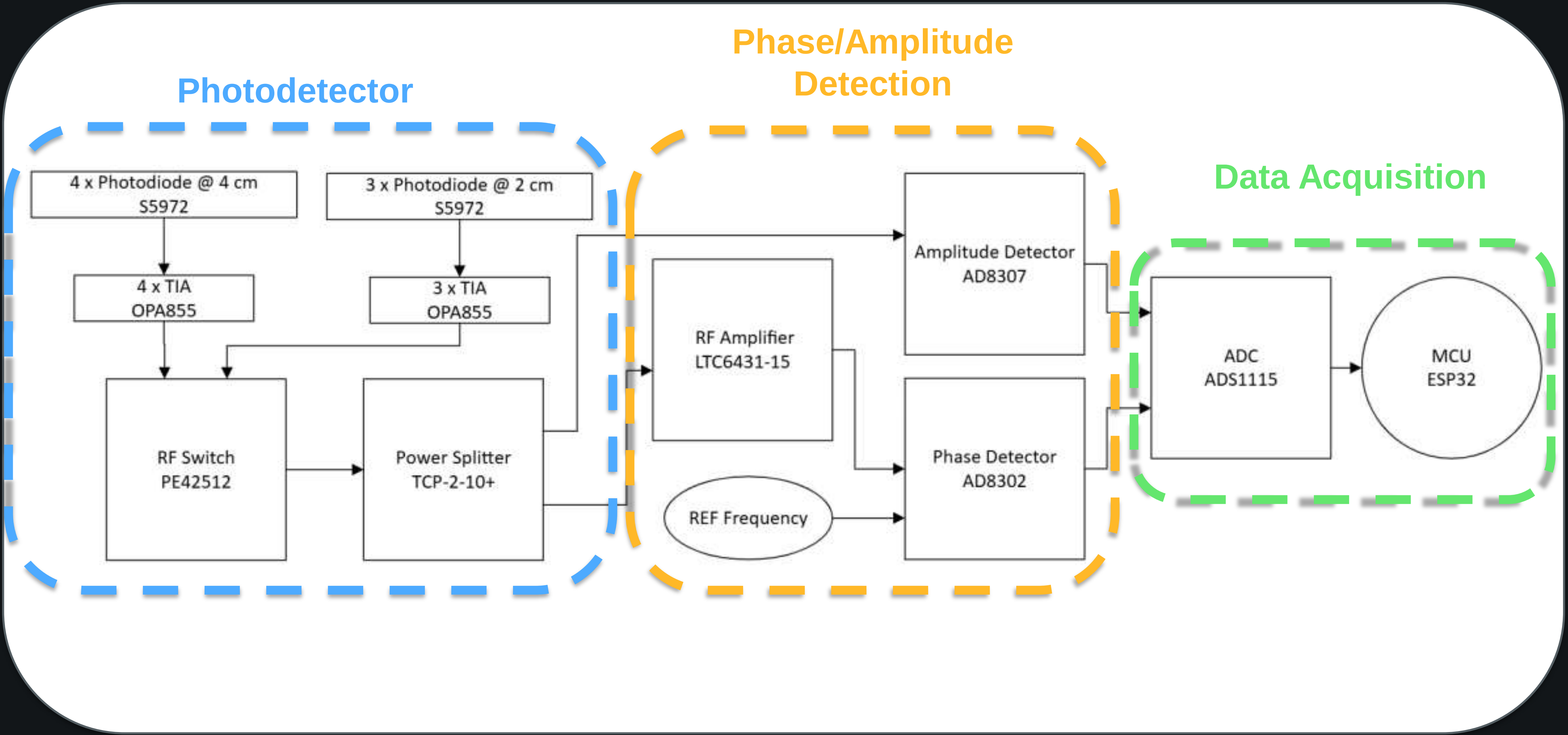


# Transmitter Subsystem



# Receiver Subsystem






Photodetector: Photodiodes + TIA + RF Switch + Splitter

Phase/Amplitude Detector: RF Amp, detectors, REF frequency

Data Acquisition: ADC + ESP32

# Transmitter Subsystem Calculation method and Formulas



## Transmitter Subsystem Formula Map

### RF Modulation

$$\Delta P_{\mathrm{opt},\lambda} = \eta_{L,\lambda} I_{\mathrm{RF},\lambda}$$

$$\omega = 2\pi f$$

$$I_{\mathrm{laser},\lambda}(t) = I_{\mathrm{bias},\lambda} + I_{\mathrm{RF},\lambda}\cos(\omega t)$$

Sets the modulation frequency and RF current swing.

### DC Bias / Linearity

$$\eta = \frac{P_{\mathrm{optical}}}{P_{\mathrm{elec}}} \times 100\%$$

$$I_{\mathrm{bias},\lambda} - I_{\mathrm{RF},\lambda} > I_{\mathrm{th},\lambda}$$

$$P_{\mathrm{elec},\lambda} = V_{F,\lambda} I_{F,\lambda}$$

Checks that the laser stays above threshold and in the linear modulation region.

### Laser Output Stage

$$P_{\mathrm{dBm}} = 10\log_{10}(P_{\mathrm{mW}})$$

$$\Delta P_{\mathrm{opt},\lambda} = m_\lambda P_{\mathrm{DC},\lambda}$$

$$P_{\mathrm{tx},\lambda} = P_{\mathrm{DC},\lambda} + \Delta P_{\mathrm{opt},\lambda}\cos(\omega t)$$

Relates modulation depth to the optical power swing at 670, 800, and 850 nm.

# Transmitter Optical Output Calculation

| Wavelength | **Modulation Depth** | Current Swing | Limit Check | **Avg. dBm** | **Ptx (t) at 500 MHz** |
|---|---|---|---|---|---|
| 670nm | 20% | 4.50–6.00 mA | 3.5<I<7 mA | 0 dBm | 1.5+0.30cos(ωt) mW |
| 800nm | 20% | 4.50–5.50 mA | 4<I<8 mA | −3.01 dBm | 0.5+0.10cos(ωt) mW |
| 850nm | 50% | 4.00–9.00 mA | 1.5<I<12 mA | 3.01 dBm | 2.0+1.00cos(ωt) mW |

ω=3.142×109 rad/s - 500MHz

# Electrical Power & Efficiency Check



| **Wavelength** | VF | **IF** | Pelec =VF IF | Optical efficiency |
|---|---|---|---|---|
| 670nm | 2.1 V | 4 mA | 8.4mW | 11.9% |
| 800nm | 1.9 V | 5.0 mA | 9.5 mW | 5.3% |
| 850nm | 1.8 V | 6.5 mA | 11.7 mW | 17.1% |

# Transmitter Calculations — Optical Launch to Tissue



## Transmitter Calculations Optical Launch to Tissue

### Optical Source Intensity

$$I_{0,\lambda} = \frac{P_{\mathrm{tx},\lambda}}{A_{\mathrm{beam},\lambda}}$$

Baseline optical intensity launched into tissue.

### Effective Attenuation

$$\mu_{\mathrm{eff},\lambda} = \sqrt{3\mu_{a,\lambda}(\mu_{a,\lambda} + \mu'_{s,\lambda})}$$

Combines absorption and scattering into intensity loss.

### Tissue Propagation

$$I_\lambda = I_{0,\lambda} e^{-\mu_{\mathrm{eff},\lambda} d}$$

Intensity decreases through forehead and brain tissue.

### Receiver Input

$$P_{\mathrm{det},\lambda} \propto I_\lambda$$

The detector input follows the tissue-reduced intensity.

# Receiver Subsystem — Signal Chain & Formula Map

## Receiver Subsystem Signal Chain & Formula Map

### Updated transmitter input

$$P_{tx,\lambda}(t) = P_{DC,\lambda} + \Delta P_{opt,\lambda}\cos(\omega t)$$

### Detected power split

$$P_{det,DC,\lambda} = P_{DC,\lambda} e^{-\mu_{eff,\lambda} d}$$

$$\Delta P_{det,\lambda} = \Delta P_{opt,\lambda} e^{-\mu_{eff,\lambda} d}$$

### DC path

$$P_{DC,\lambda} \rightarrow P_{det,DC,\lambda} \rightarrow I_{avg,\lambda} \rightarrow P_{noise,\lambda}$$

$$I_{avg,\lambda} = R_{D,\lambda} P_{det,DC,\lambda}$$

Used for average photodiode current and receiver noise.

### AC modulation path

$$\Delta P_{opt,\lambda} \rightarrow \Delta P_{det,\lambda} \rightarrow V_{AC,\lambda} \rightarrow P_{sig,\lambda} \rightarrow SNR_{\lambda}$$

$$I_{AC,\lambda} = R_{D,\lambda} \Delta P_{det,\lambda}$$

Used for recovered modulation signal, IQ amplitude, signal power, and SNR.

# Receiver Calculations — Noise Power & SNR

## Receiver Calculations Noise Power & SNR

### Noise Current Terms

$$i^2_{\text{shot},\lambda} = 2qI_{\text{avg},\lambda}B$$

$$i^2_{\text{RIN},\lambda} = RIN \cdot I^2_{\text{avg},\lambda}B$$

$$P_{\text{thermal}} = k_B T B$$

Noise grows with current, bandwidth, and electronics.

### Total Receiver Noise

$$i_{\text{noise},\lambda} = \sqrt{i^2_{\text{shot}} + i^2_{\text{RIN}} + i^2_{\text{thermal}}}$$

$$P_{\text{noise},\lambda} = \frac{(i_{\text{noise},\lambda} Z_T)^2}{2R_L}$$

Converts total current noise into receiver noise power.

### Signal Reliability

$$SNR_\lambda = \frac{P_{\text{sig},\lambda}}{P_{\text{noise},\lambda}}$$

$$P_{\text{sig},\lambda} > P_{\text{noise},\lambda}$$

Main metric: signal reliability and clear recovery.

# IQ Demodulation & Signal Recovery

## IQ Demodulation & Signal Recovery

### Received RF Signal

$$s_\lambda(t) = A_\lambda \cos(\omega t + \phi_\lambda)$$

$$\omega = 2\pi f$$

Detected optical signal keeps amplitude and phase after tissue propagation.

### I/Q Recovery

$$I_\lambda = \mathrm{LPF}\{s_\lambda(t)\cos(\omega t)\}$$

$$Q_\lambda = \mathrm{LPF}\{s_\lambda(t)\sin(\omega t)\}$$

$$A_\lambda = \sqrt{I_\lambda^2 + Q_\lambda^2}$$

$$\phi_\lambda = \tan^{-1}\left(\frac{Q_\lambda}{I_\lambda}\right)$$

Demodulation extracts baseband components and reconstructs amplitude and phase.

### How to Use Them

$$I_{\mathrm{rec},\lambda} \propto A_\lambda$$

Amplitude is used for intensity and MBLL. Phase is handled separately for delay and path information.

# Calculations — Intensity-Based MBLL



## Calculations Intensity-Based MBLL

### Optical Density from Intensity

$$\Delta OD_\lambda = \log_{10}\left(\frac{I_{0,\lambda}}{I_\lambda}\right)$$

I0 is baseline intensity; Iλ is measured received intensity.

### Three-Wavelength Measurement

$$\lambda = 670\ \text{nm},\ 800\ \text{nm},\ 850\ \text{nm}$$

Each wavelength gives a separate optical-density equation.

### Modified Beer–Lambert Law

$$\Delta OD_\lambda = \varepsilon_{\text{HbO}_2,\lambda}\Delta[\text{HbO}_2]L_{\text{eff},\lambda} + \varepsilon_{\text{Hb},\lambda}\Delta[\text{Hb}]L_{\text{eff},\lambda}$$

MBLL is based on intensity change, not phase directly.

### Matrix / Output

$$\Delta OD = E\Delta C L_{\text{eff}}$$

$$\Delta[\text{HbO}_2],\ \Delta[\text{Hb}]$$

Outputs: oxygenated and deoxygenated hemoglobin changes.

# Calculations & Diagram

## Calculations & Diagram

### Wavelength Set

$$\lambda = 670,\ 800,\ 850\ \mathrm{nm}$$

Selected wavelengths sample different hemoglobin responses.

### Intensity → Optical Density

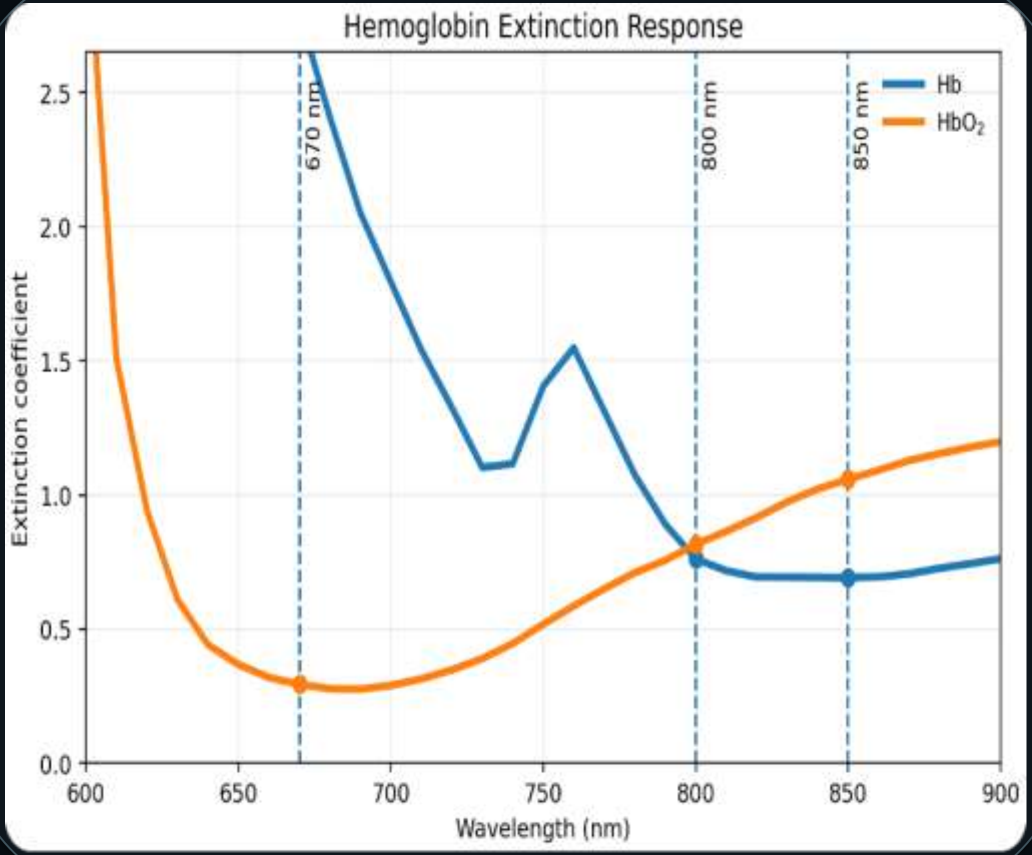
$$\Delta OD_\lambda = \log_{10}\left(\frac{I_{0,\lambda}}{I_\lambda}\right)$$

Each wavelength produces an optical-density change.

### MBLL Estimation

$$\Delta OD_\lambda \rightarrow \Delta[HbO_2],\ \Delta[Hb]$$

Relative intensity changes across wavelengths support screening interpretation.




*Hb and $HbO_2$ response curves with 670, 800, and 850 nm marked.*

# Detailed Calculations - Tissue Depth, Loss, and Leakage Location Clue

## Core equations

$$\mu_{\mathrm{eff},\lambda} = \sqrt{3\mu_{a,\lambda}(\mu_{a,\lambda} + \mu'_{s,\lambda})}$$

$$L_{\mathrm{tissue,\,dB},\lambda} = 4.343\,\mu_{\mathrm{eff},\lambda} d$$

$$\Delta OD_{\lambda,d} = \log_{10}\left(\frac{I_{0,\lambda,d}}{I_{\lambda,d}}\right)$$

$$\uparrow \mu_a \Rightarrow \uparrow \mu_{\mathrm{eff}} \Rightarrow \uparrow L_{\mathrm{tissue}} \Rightarrow \downarrow I_\lambda$$

## d1 / d2 table for μa and μ′s

| Path | d | μa estimate | μ′s estimate | Use |
|---|---|---|---|---|
| d1 | 2 cm | from ΔODλ,d1 | article/model fit | shallow clue |
| d2 | 4 cm | from ΔODλ,d2 | article/model fit | deeper clue |

## Loss for two depths

| λ | μeff | L @ d1=2 cm | L @ d2=4 cm | Gain |
|---|---|---|---|---|
| 670 | 2.31 | 20.1 dB | 40.1 dB | +20.1 dB |
| 800 | 2.04 | 17.7 dB | 35.4 dB | +17.7 dB |
| 850 | 2.17 | 18.8 dB | 37.7 dB | +18.8 dB |

*Gain = less loss using the shallow 2 cm path instead of the deeper 4 cm path.*

## Leakage location clue

- Blood leakage can increase local absorption μa.
- Higher μa increases μeff and lowers detected intensity.
- Large ΔOD at d1 suggests a shallower absorption change.
- Large ΔOD at d2 suggests a deeper path is affected.
- Use 670, 800, 850 nm together to check hemoglobin response.

# Detailed Calculations - Receiver Current, Voltage & Signal Power



## Formula used

$$P_{det,DC,\lambda} = P_{DC,\lambda} e^{-\mu_{eff,\lambda} d}$$

$$\Delta P_{det,\lambda} = \Delta P_{opt,\lambda} e^{-\mu_{eff,\lambda} d}$$

The DC part sets average receiver level; the AC part sets recovered modulation signal.

## d = 2 cm

| λ | Pdet,DC | ΔPdet |
|---|---|---|
| 670 | 14.779 µW | 2.956 µW |
| 800 | 8.454 µW | 1.691 µW |
| 850 | 26.073 µW | 13.037 µW |

## d = 4 cm

| λ | Pdet,DC | ΔPdet |
|---|---|---|
| 670 | 0.146 µW | 0.029 µW |
| 800 | 0.143 µW | 0.029 µW |
| 850 | 0.340 µW | 0.170 µW |

## Meaning

- d = 2 cm gives stronger receiver input.
- d = 4 cm probes deeper but has much more attenuation.
- ΔPdet is the modulated signal that supports IQ and SNR.

# Detailed Calculations - IQ Demodulation with d = 2 cm and 4 cm

**Receiver equations**

$$I_{avg,\lambda} = R_D P_{det,DC,\lambda}$$

$$I_{AC,\lambda} = R_D \Delta P_{det,\lambda}$$

$$V_{out,DC} = I_{avg} Z_T, \quad V_{AC} = I_{AC} Z_T$$

$$P_{sig,\lambda} = \frac{V_{AC,\lambda}^2}{2R_L}$$

Constants: RD = 0.65 A/W, ZT = 794 Ω, RL = 50 Ω

## d = 2 cm

| λ | Iavg | Vout,DC | IAC | VAC | Psig |
|---|---|---|---|---|---|
| 670 | 9.606 µA | 7.628 mV | 1.921 µA | 1.526 mV | −46.33 dBm |
| 800 | 5.495 µA | 4.363 mV | 1.099 µA | 0.873 mV | −51.18 dBm |
| 850 | 16.947 µA | 13.456 mV | 8.474 µA | 6.728 mV | −33.44 dBm |

## d = 4 cm

| λ | Iavg | Vout,DC | IAC | VAC | Psig |
|---|---|---|---|---|---|
| 670 | 0.095 µA | 75.15 µV | 0.019 µA | 15.03 µV | −86.46 dBm |
| 800 | 0.093 µA | 73.77 µV | 0.019 µA | 14.75 µV | −86.62 dBm |
| 850 | 0.221 µA | 175.42 µV | 0.110 µA | 87.71 µV | −71.14 dBm |

# Detailed Calculations - Intensity-Based MBLL



## Optical density from intensity

$$\Delta OD_\lambda = \log_{10}\left(\frac{I_{0,\lambda}}{I_\lambda}\right)$$

## Modified Beer-Lambert Law

$$\Delta OD_\lambda = \varepsilon_{\mathrm{HbO_2},\lambda}\Delta[\mathrm{HbO2}]L_{\mathrm{eff},\lambda} + \varepsilon_{\mathrm{Hb},\lambda}\Delta[\mathrm{Hb}]L_{\mathrm{eff},\lambda}$$

## Matrix form

$$\Delta OD = E \cdot \Delta C \cdot L_{\mathrm{eff}}$$

## Extinction coefficient table

| λ | $\varepsilon HbO_2$ | εHb |
|---|---|---|
| 670 | 0.000294 | 0.002795 |
| 800 | 0.000816 | 0.000762 |
| 850 | 0.001058 | 0.000691 |

## Expected output

- $\Delta[HbO_2]$
- Δ[Hb]
- Oxygenation trend from intensity changes.

# Detailed Calculations - Noise Power & SNR



## SNR equation

$$SNR_{dB} = P_{sig,\,dBm} - P_{noise,\,dBm}$$

The signal power uses the AC recovered modulation signal from ΔPdet.

### d = 2 cm

| λ | Pnoise | Psig | SNR |
|---|---|---|---|
| 670 | −135.95 dBm | −46.33 dBm | 89.62 dB |
| 800 | −138.00 dBm | −51.18 dBm | 86.82 dB |
| 850 | −133.56 dBm | −33.44 dBm | 100.12 dB |

### d = 4 cm

| λ | Pnoise | Psig | SNR |
|---|---|---|---|
| 670 | −143.63 dBm | −86.46 dBm | 57.17 dB |
| 800 | −143.64 dBm | −86.62 dBm | 57.02 dB |
| 850 | −143.38 dBm | −71.14 dBm | 72.24 dB |

### Meaning

- Higher SNR supports clearer intensity recovery.
- 2 cm has higher SNR because less tissue loss occurs.
- 4 cm has lower SNR but gives deeper-path information.

# Conclusion

- Portable FD-fNIR system designed for early TBI screening.

- Intended as a real-time screening tool, not a replacement for CT or MRI.

**Tx** Transmitter — **Brain Tissue** Channel — **Rx** Receiver

## Thank You

### Questions?

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# Detailed Calculations - Transmitter Frequency & Optical Power

## Modulation frequency

$$\omega = 2\pi f = 2\pi(500 \times 10^6) = 3.142 \times 10^9 \text{ rad/s}$$

## Optical output power conversion

$$P_{\mathrm{dBm}} = 10\log_{10}(P_{\mathrm{mW}})$$

$$670 \text{ nm}: \; 10\log_{10}(5.0) = 6.99 \text{ dBm}$$

$$800 \text{ nm}: \; 10\log_{10}(1.65) = 2.17 \text{ dBm}$$

$$850 \text{ nm}: \; 10\log_{10}(1.85) = 2.67 \text{ dBm}$$

## Calculated transmitter optical power

| λ | Popt | Popt dBm |
|---|---|---|
| 670 nm | 5.0 mW | 6.99 dBm |
| 800 nm | 1.65 mW | 2.17 dBm |
| 850 nm | 1.85 mW | 2.67 dBm |

## Meaning

- dBm conversion helps compare optical power with dB losses.
- This supports the transmitter-to-receiver link budget.

# Detailed Calculations - RF Current & Laser Linearity

## RF modulation current

$$I_{\mathrm{RF}} = \sqrt{\frac{2P_{\mathrm{RF}}}{R_F}}, \quad P_{\mathrm{RF}} = 1.0\ \mathrm{mW}$$

$$R_F = 5\Omega: \; I_{\mathrm{RF}} = \sqrt{\frac{2(1.0 \times 10^{-3})}{5}} = 20.0\ \mathrm{mA}$$

$$R_F = 10\Omega: \; I_{\mathrm{RF}} = \sqrt{\frac{2(1.0 \times 10^{-3})}{10}} = 14.1\ \mathrm{mA}$$

**RF current range: 14.1–20.0 mA**

## Laser current and linearity

$$I_{\mathrm{laser},\lambda} = I_{\mathrm{bias},\lambda} + I_{\mathrm{RF},\lambda}\cos(\omega t)$$

$$I_{\mathrm{bias},\lambda} - I_{\mathrm{RF},\lambda} > I_{\mathrm{th},\lambda}$$

## Numbers to check

- PRF = 1.0 mW
- RF = 5–10 Ω
- IRF = 14.1–20.0 mA
- Ibias and Ith come from the final laser datasheet

## Meaning

- RF current controls the strength of intensity modulation.
- Bias condition prevents clipping and threshold drop-out.

# Detailed Calculations - Transmitter Electrical Power & Efficiency

## Electrical power

$$P_{\text{elec}} = V_F I_F$$

- Use final datasheet values for VF and IF.
- Expected electrical power stays in the mW range.

## Laser efficiency

$$\eta = \frac{P_{\text{optical}}}{P_{\text{elec}}} \times 100\%$$

$$670\text{ nm}: \ \eta = \frac{5.0}{25.0} \times 100\% = 20.0\%$$

$$800\text{ nm}: \ \eta = \frac{1.65}{20.0} \times 100\% = 8.25\%$$

$$800\text{ nm}: \ \eta = \frac{1.65}{13.6} \times 100\% = 12.1\%$$

$$850\text{ nm}: \ \eta = \frac{1.85}{17.6} \times 100\% = 10.5\%$$

$$850\text{ nm}: \ \eta = \frac{1.85}{13.6} \times 100\% = 13.6\%$$

## Expected values

| λ | Popt | Pelec | η |
|---|---|---|---|
| 670 | 5.0 mW | 25.0 mW | 20.0% |
| 800 | 1.65 mW | 13.6-20.0 mW | 8.25-12.1% |
| 850 | 1.85 mW | 13.6-17.6 mW | 10.5-13.6% |

## Meaning

- Electrical power estimates laser drive requirements.
- Efficiency shows how much electrical input becomes optical output.

# Detailed Calculations - Optical Modulation & Transmitted Power

## Optical modulation depth

$$\Delta P_{\mathrm{opt},\lambda} = \eta_{L,\lambda} I_{\mathrm{RF},\lambda}$$

- Plug in IRF = 14.1–20.0 mA.
- Plug in ηL from each selected laser diode datasheet.

## Transmitted optical waveform

$$P_{\mathrm{tx},\lambda} = P_{\mathrm{DC},\lambda} + \Delta P_{\mathrm{opt},\lambda} \cos(\omega t)$$

- PDC,670 = 5.0 mW
- PDC,800 = 1.65 mW
- PDC,850 = 1.85 mW

## Expected output

- A DC optical output plus RF optical modulation.
- This modulated light enters the forehead/brain tissue.
- Exact ΔPopt needs final slope efficiency.

## Meaning

- This connects transmitter hardware to the optical signal.
- The receiver later detects the attenuated version of this signal.